

Assignment 1

Course: 24ECJ502-Digital Signal Processing

Platform: Google Colab (Python-based Implementation)

Target: Field-based/Experiential Learning based assignment.

Mapping - POs : PO1,PO2, PO5, PO6,PO9,PO11

Wks : WK1,WK2,WK3,WK4, WK5, WK6,WK7,WK8

Sustainability: SDG 3, SDG 9, SDG 12

Assignment Title:

“Spectral Analysis and Basic Frequency-Domain Filtering of Real-World Audio using DFT (FFT)”

Context:

You are working as a signal processing engineer exploring real-world audio environments for applications like smart hearing aids, classroom acoustics monitoring, or public announcement systems. In this assignment, you will record two short audio clips of your own voice in different environments, perform DFT to analyze their frequency content, and attempt basic noise reduction using a simple zeroing technique in the frequency domain.

Learning Objectives:

- Understand and apply DFT(FFT) to real audio signals
 - Visualize and interpret frequency-domain plots
 - Perform elementary filtering by zeroing frequency bins
 - Reflect on signal processing applications in the context of sustainable development
 - Practice report writing and effective communication of engineering results
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Tasks for Students:

① *Record and Upload Audio*

- Record a **10-second audio clip** (WAV format, 16 kHz or above) containing one from a closed, quiet room and one from a noisy environment (e.g., classroom, hostel, campus, street)

- Upload it to your Colab environment.

② *Preprocessing*

- Normalize and trim the audio signals
- Plot the **time-domain waveform**.

③ *FFT Analysis*

- Compute the DFT (FFT)
- Plot and compare their **magnitude spectrum**.
- Identify dominant frequency regions and discuss how noise affects the spectrum.

④ *DFT (FFT)-Based Filtering (Zeroing Technique)*

- Zero out unwanted frequency bins (e.g., below 100 Hz or above 3 kHz) from the noisy audio.
- Perform inverse DFT(FFT).
- Normalize and playback filtered audio.
- Plot and compare waveforms and spectra before and after filtering.

⑤ *Reflection on Sustainable Goals*

Write a **100–150 word reflection** on how real-world signal analysis can support any one **UN Sustainable Development Goal (SDG)** such as:

- SDG 3: Developing assistive hearing tech for better well-being
- SDG 9: Enhancing acoustic sensing for smart infrastructure
- SDG 12: Promoting responsible and efficient audio processing

6 *Communication and Reporting*

Prepare a concise and clear report (4–5 pages) that:

- Summarizes your methods and findings
- Contains screenshots/graphs of all plots
- Compares clean vs noisy signal behavior
- Documents your zeroing/filtering outcome
- Includes the SDG reflection

Evaluation Rubric (Total: 30 Marks)

Criteria	Excellent (5)	Good (3)	Poor (1)
Audio Sample Quality	Clear, relevant noise context	Minor noise issues	Incomplete or irrelevant
FFT Code & Plot	Accurate, labeled frequency-domain plot	Minor code/labeling	Incorrect or missing plot

Criteria	Excellent (5)	Good (3)	Poor (1)
Frequency Analysis	Meaningful and correct interpretation	Vague but mostly correct issues	Incorrect or absent
SDG Reflection	Relevant, insightful application	Moderately connected	Weak or missing
FFT Filtering via Zeroing	Audible impact, visually clear result	Works but minimal effect	Ineffective or incorrect
Report Quality & Communication	Clear, structured, well-written	Adequate explanation	Poorly presented or unclear

Tools Suggested:

- Python libraries: numpy, scipy, matplotlib, librosa, IPython.display.Audio
 - Platform: Google Colab (free and shareable)
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Hints:

- Apply windowing before FFT to reduce leakage.
 - Try changing your masking threshold to hear the effect.
 - Always normalize signals before playback.
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✦✦ Last Date of Submission

25-07-2026
